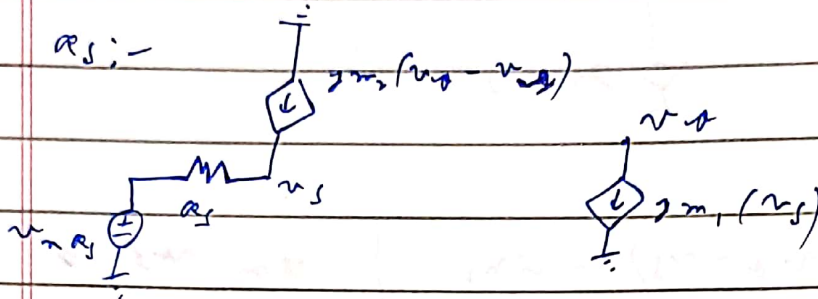


Tutorial - 2

2.4.20

A1. R_S :-

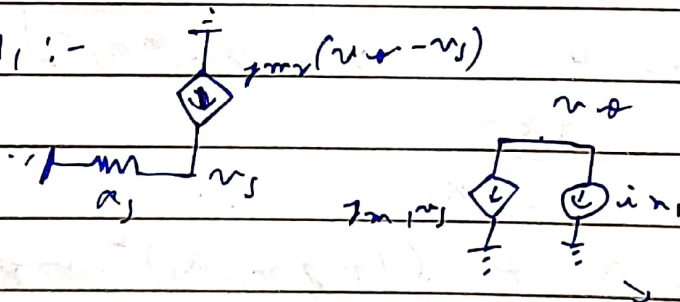
$$v_s - v_{gs} = g_{m2} R_S (v_g - v_s)$$

$$v_g = \frac{v_s (g_{m2} R_S + 1) - v_{gs}}{g_{m2} R_S}$$

$$v_s = 0$$

$$\therefore g_{m1} v_s = 0$$

$$\therefore v_g = -\frac{v_{gs}}{g_{m2} R_S} \Rightarrow S_{v_{gs}} = \frac{4kT}{g_{m2}^2 R_S^2} = \frac{4kT}{g_{m2}^2 R_S}$$

A2. M_1 :-

$$g_{m2} (v_g - v_s) = \frac{v_s}{R_S}$$

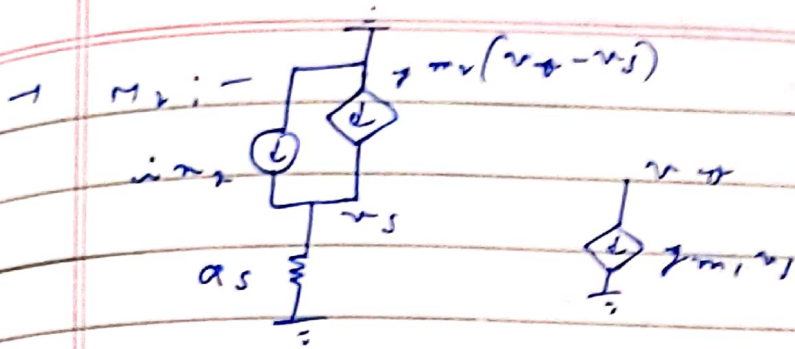
$$g_{m1} v_s + i_{n1} = 0$$

$$v_s = -\frac{i_{n1}}{g_{m1}}$$

$$g_{m2} v_g = v_s \left(\frac{1}{R_S} + g_{m2} \right)$$

$$v_g = v_s \left(\frac{1 + g_{m2} R_S}{g_{m2} R_S} \right) = -\frac{i_{n1}}{g_{m1}} \left(\frac{1 + g_{m2} R_S}{g_{m2} R_S} \right)$$

$$\therefore S_{v_{gs}} = \frac{4kT}{g_{m1}} \left(\frac{1 + g_{m2} R_S}{g_{m2} R_S} \right)^2$$



$$\frac{v_s}{R_S} = g_{m2}(v_{gs} - v_s) + i_{n2} \quad \left| \quad g_{m1} v_s = 0 \right.$$

$$\therefore v_s = 0$$

$$\therefore v_{gs} = - \frac{i_{n2}}{g_{m2}}$$

$$\therefore S v_{n\theta} = \frac{4kT}{g_{m2}}$$

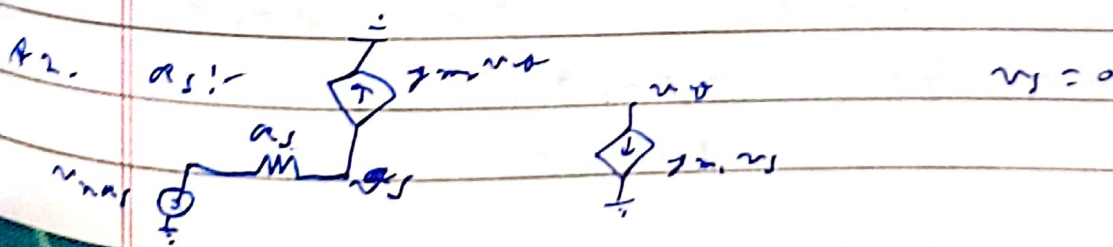
$$\rightarrow f = 1 + \frac{4kT}{g_{m2}} \left[1 + \frac{(1 + g_{m2} R_S)^2}{g_{m1} g_{m2} R_S^2} \right]$$

$$\frac{4kT}{g_{m2}^2 R_S}$$

$$= 1 + g_{m2} R_S + \frac{(1 + g_{m2} R_S)^2}{g_{m1} R_S^2}$$

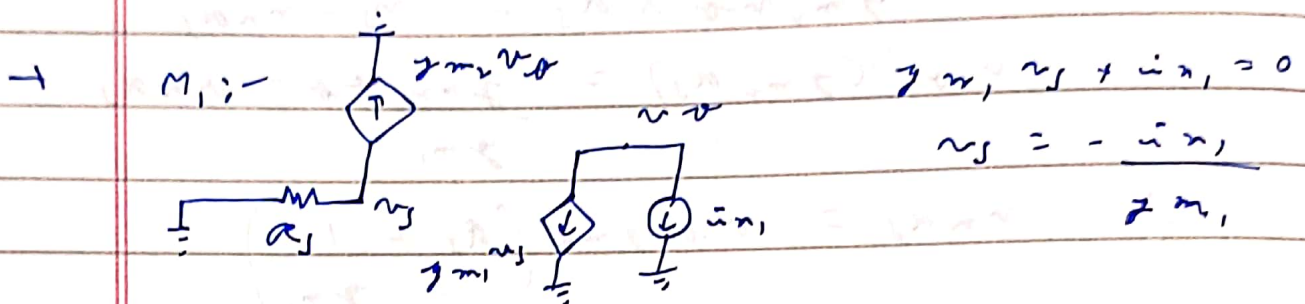
$$f = (1 + g_{m2} R_S) \left[1 + \frac{1 + g_{m2} R_S}{g_{m1} R_S} \right]$$

$$f = \frac{(1 + g_{m2} R_S) [1 + (g_{m1} + g_{m2}) R_S]}{g_{m1} R_S} \quad \left| \quad NF = 10 \log_{10} f \right.$$



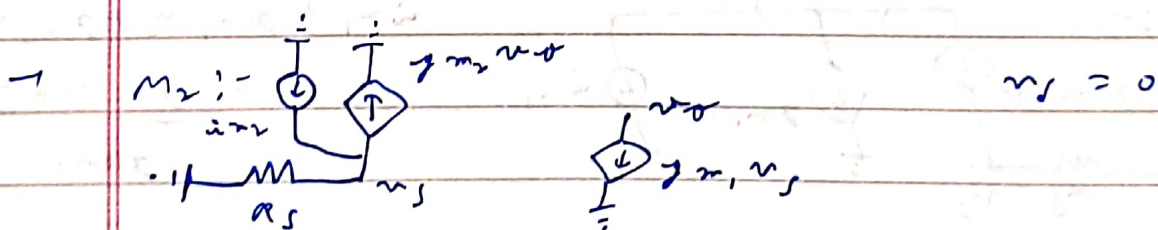
$$\frac{v_{ns} - v_s}{\alpha_s} = g_{m2} v_d \Rightarrow v_d = \frac{v_{ns} \alpha_s}{g_{m2} \alpha_s}$$

$$\therefore S_{vnd} = \frac{4kT}{g_{m2}^2 \alpha_s}$$



$$\frac{0 - v_s}{\alpha_s} = g_{m2} v_d \Rightarrow v_d = \frac{-v_s}{g_{m2} \alpha_s} = \frac{i_{n1}}{g_{m2} g_{m2} \alpha_s}$$

$$\therefore S_{vnd} = \frac{4kT}{g_{m1} (g_{m2} \alpha_s)^2}$$



$$\therefore i_{n2} = g_{m2} v_d \Rightarrow S_{vnd} = \frac{S_{i_{n2}}}{g_{m2}^2} = \frac{4kT}{g_{m2}}$$

→

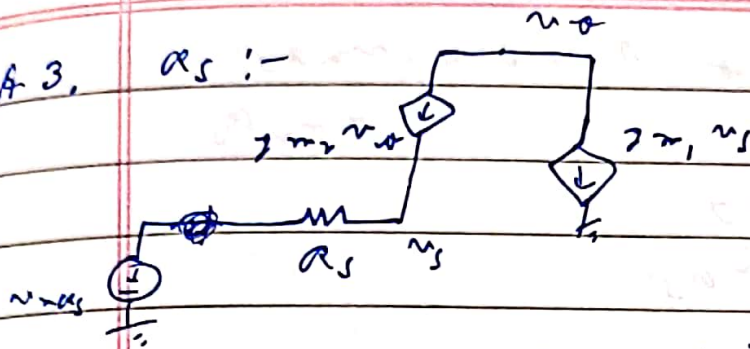
$$F = 1 + \frac{4kT}{g_{m2}} \left[1 + \frac{1}{g_{m1} g_{m2} \alpha_s^2} \right] = 1 + g_{m2} \alpha_s + \frac{1}{g_{m1} \alpha_s}$$

$\frac{4kT}{g_{m2}^2 \alpha_s}$

$NF = 10 \log_{10} F$

A3.

$\alpha_s :-$



$$g_{m1} v_s + g_{m2} v_o = 0$$

$$\therefore v_s = -\frac{g_{m2} v_o}{g_{m1}}$$

$$g_{m2} v_o \times R_s = v_s - v_o \alpha_s$$

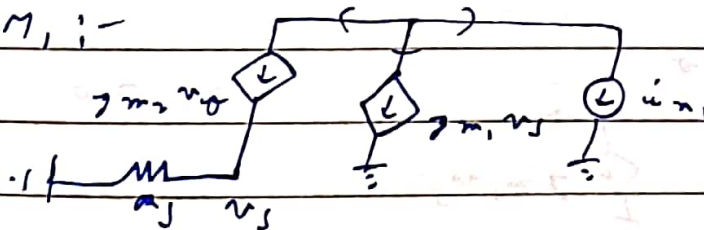
$$\therefore v_o (g_{m2} R_s) = -\frac{g_{m2} v_o}{g_{m1}} - v_o \alpha_s$$

$$\therefore -v_o \alpha_s = v_o \frac{g_{m2}}{g_{m1}} (R_s + 1)$$

$$\therefore v_o = -v_{na1} \times \frac{g_{m1}}{g_{m2} (1 + g_{m1} R_s)}$$

$$\therefore S v_{na1} = \frac{4 \times 7 R_s \times g_{m1}^2}{g_{m2}^2 (1 + g_{m1} R_s)^2}$$

M1 :-



$$i_{n1} + g_{m1} v_s + g_{m2} v_o = 0$$

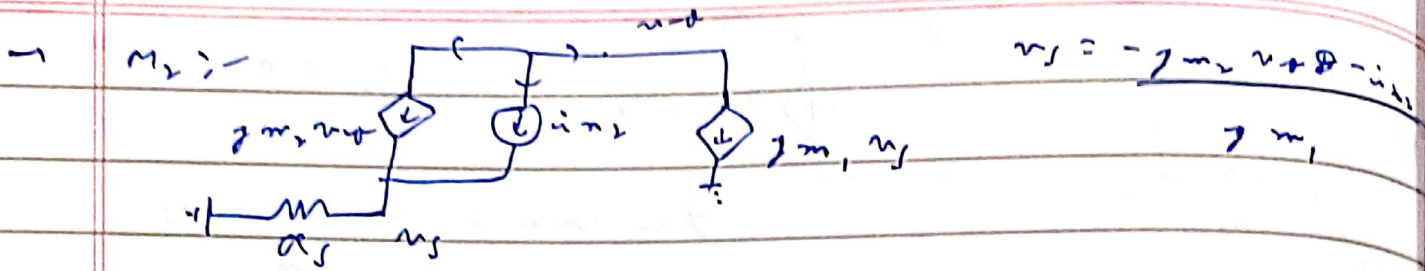
$$\therefore v_s = -\frac{g_{m2} v_o + i_{n1}}{g_{m1}}$$

$$\alpha_s g_{m2} v_o = v_s = -\frac{g_{m2} v_o + i_{n1}}{g_{m1}}$$

$$v_o (\alpha_s g_{m1} + 1) g_{m2} = -i_{n1}$$

$$\therefore S v_{na1} = \frac{4 \times 7 g_{m1}}{g_{m2}^2 (1 + \alpha_s g_{m1})^2}$$

M2 :-



$$\therefore g_{m2} v_{gs} + i_{r2} = \frac{v_s}{\alpha_s} = -\frac{(g_{m2} v_{gs} + i_{r2})}{g_{m1} \alpha_s}$$

$$\therefore (g_{m2} v_{gs} + i_{r2}) \left(1 + \frac{1}{g_{m1} \alpha_s} \right) = 0$$

$$\therefore v_{gs} = -\frac{i_{r2}}{g_{m2}} \Rightarrow \text{S.V.T} = \frac{4 \times 7}{g_{m2}}$$

$$1. \quad F = 1 + \frac{4 \times 7}{g_{m2}} \left[1 + \frac{g_{m1}}{g_{m2} (1 + \alpha_s g_{m1})^2} \right]$$

$$\frac{4 \times 7 \alpha_s \times g_{m1}^2}{g_{m2}^2 (1 + g_{m1} \alpha_s)^2}$$

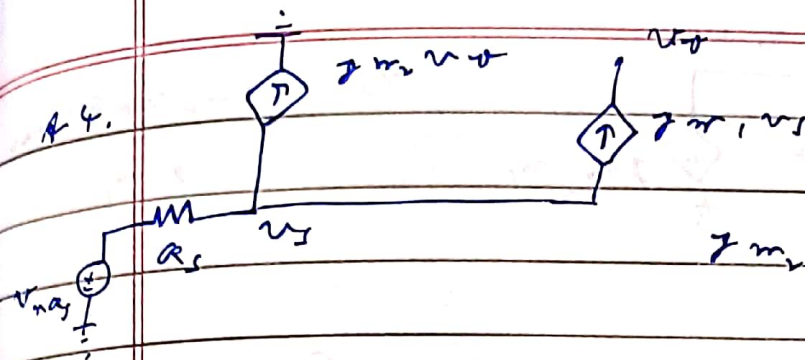
$$= 1 + \frac{g_{m2} (1 + g_{m1} \alpha_s)^2}{g_{m1}^2 \alpha_s} \left[1 + \frac{g_{m1}}{g_{m2} (1 + \alpha_s g_{m1})^2} \right]$$

$$= 1 + \frac{g_{m2} (1 + g_{m1} \alpha_s)^2}{g_{m1}^2 \alpha_s} + \frac{1}{g_{m1} \alpha_s}$$

$$= \frac{g_{m1}^2 \alpha_s + g_{m1} + g_{m2} (1 + g_{m1} \alpha_s)}{g_{m1}^2 \alpha_s}$$

$$F = \frac{(g_{m1} + g_{m2}) (1 + g_{m1} \alpha_s)}{g_{m1}^2 \alpha_s}$$

$$N F = 10 \log_{10} F$$



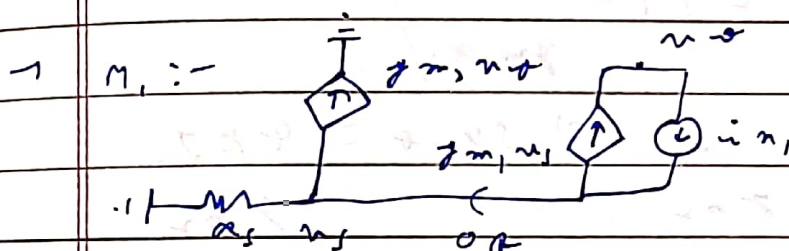
$$v_s = 0$$

$$g_{m1} v_o = \frac{v_o R_s - v_s}{R_s}$$

$$\therefore v_o = \frac{v_o R_s}{g_{m1} R_s} \Rightarrow 5 v_o = 4 R_s$$

$$g_{m1} R_s$$

$$g_{m1}^2 R_s$$



$$g_{m1} v_s = i_{sc}$$

$$v_s = i_{sc} R_s$$

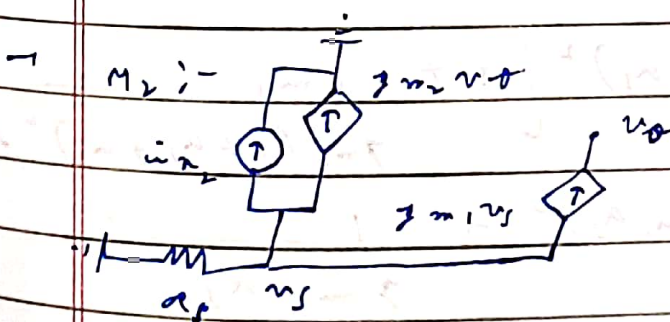
$$g_{m1}$$

$$0 - v_s = \frac{g_{m2} v_o}{g_{m1}} \Rightarrow v_o = -i_{sc} R_s$$

$$g_{m1} g_{m2} R_s$$

$$\therefore 5 v_o = 4 R_s$$

$$g_{m1} (g_{m2} R_s)^2$$



$$v_s = 0$$

$$\therefore g_{m1} v_s + i_{sc} R_s = 0$$

$$\therefore 5 v_o = \frac{5 i_{sc} R_s}{g_{m1}} = \frac{4 R_s}{g_{m1}}$$

$$g_{m1}^2 R_s$$

$$g_{m2}$$

$$F = 1 + \frac{4 R_s}{g_{m1}} \left[1 + \frac{1}{g_{m1} g_{m2} R_s^2} \right] = 1 + g_{m1} R_s + \frac{1}{g_{m1} R_s}$$

$$4 R_s$$

$$g_{m1}^2 R_s$$

$$\therefore \text{NF} = 10 \log_{10} F$$

A 5. $NF = 3dB \Rightarrow F = 10^{0.3} = \frac{\text{tot. noise } P}{\text{noise } P, a_s}$

$$\text{noise } P, a_s = \frac{x}{100} \times \text{tot. noise } P$$

$$\therefore \frac{x}{100} = \frac{1}{10^{0.3}} \Rightarrow x = 10^{1.7} = 50.12\%$$

A 6. $NF = 1dB \Rightarrow F = 10^{0.1} = \frac{\text{tot. w/f noise } P}{\text{w/f noise } P, a_s}$

$$\therefore \frac{x}{100} = \frac{1}{10^{0.1}} \Rightarrow x = 10^{1.9} = 79.43\%$$